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→ For each of the systems described by the following diff. eq., find the s/m transfer fn.

$$(i) \frac{d^2 y(t)}{dt^2} + 11 \frac{dy(t)}{dt} + 24 y(t) = 5 \frac{dx(t)}{dt} + 3x(t)$$

$$(ii) \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} \quad \text{if } y(0^-) = 2 \\ \frac{dy(0^-)}{dt} = 1$$

$$\text{and } x(t) = e^{-t} u(t)$$

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$$A: (i) \frac{d^2 y(t)}{dt^2} + 11 \frac{dy(t)}{dt} + 24 y(t) = 5 \frac{dx(t)}{dt} + 3x(t)$$

Taking L.T OBS.

$$s^2 Y(s) + 11s Y(s) + 24 Y(s) = 5sX(s) + 3X(s)$$

$$(s^2 + 11s + 24) Y(s) = (5s + 3) X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{5s + 3}{s^2 + 11s + 24}$$

Find $y(t)$.

$$(ii) \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}$$

$$y(0^-) = 2; \frac{dy(0^-)}{dt} = 1; x(t) = e^{-t} u(t)$$

Taking L.T. O.B.S.

$$s^2 Y(s) - s y(0^-) - \frac{dy(0^-)}{dt} + 3[s Y(s) - y(0^-)] + 2Y(s) = s X(s) - x(0^-)$$

Sub $y(0^-)$ and $\frac{dy(0^-)}{dt}$

$$s^2 Y(s) - 2s - 1 + 3[s Y(s) - 2] + 2Y(s) = s X(s)$$

$$Y(s)[s^2 + 3s + 2] - 2s - 1 - 6 = s X(s)$$

$$Y(s)[s^2 + 3s + 2] - 2s - 7 = s X(s)$$

$$x(t) = e^{-t} u(t) \rightarrow X(s) = \frac{1}{s+1}$$

$$Y(s)[s^2 + 3s + 2] - 2s - 7 = \frac{s}{s+1}$$

$$\begin{aligned}
 Y(s) [s^2 + 3s + 2] &= \frac{s}{s+1} + 2s + 7 \\
 &= \frac{s + (2s+7)(s+1)}{s+1} \\
 &= \frac{s + 2s^2 + 2s + 7s + 7}{s+1}
 \end{aligned}$$

$$Y(s) (s^2 + 3s + 2) = \frac{2s^2 + 10s + 7}{s+1}$$

$$\begin{aligned}
 Y(s) &= \frac{2s^2 + 10s + 7}{(s+1)(s^2 + 3s + 2)} \\
 &= \frac{2s^2 + 10s + 7}{(s+1)(s+1)(s+2)}
 \end{aligned}$$

$$\begin{aligned}
 Y(s) &= \frac{2s^2 + 10s + 7}{(s+1)^2 (s+2)} \\
 &= \frac{A}{(s+1)^2} + \frac{B}{(s+1)} + \frac{C}{s+2}
 \end{aligned}$$

$$\begin{aligned}
 A &= \cancel{(s+1)^2} \frac{2s^2 + 10s + 7}{\cancel{(s+1)^2} (s+2)} \Big|_{s=-1} \\
 &= \frac{2 + -10 + 7}{-1 + 2} = \underline{\underline{-1}}
 \end{aligned}$$

$$B = \frac{1}{11} \frac{d}{ds} \left[\frac{(s+1)^2}{(s+1)^2(s+2)} \frac{2s^2+10s+7}{(s+1)^2(s+2)} \right] \Big|_{s=-1}$$

$$= \frac{(s+2)(4s+10) - (2s^2+10s+7) \cdot 1}{(s+2)^2} \Big|_{s=-1}$$

$$= \frac{4s^2 + 10s + 8s + 20 - 2s^2 - 10s - 7}{(s+2)^2} \Big|_{s=-1}$$

$$= \frac{2s^2 + 8s + 13}{(s+2)^2} \Big|_{s=-1}$$

$$= \frac{2 - 8 + 13}{(-1+2)^2} = \frac{7}{1} = 7$$

$$C = \frac{(s+2)}{(s+1)^2(s+2)} \frac{2s^2+10s+7}{(s+1)^2(s+2)} \Big|_{s=-2}$$

$$= \frac{2 \times 4 - 20 + 7}{(-2+1)^2} = \frac{-5}{1} = -5$$

$$Y(s) = \frac{-1}{(s+1)^2} + \frac{7}{s+1} - \frac{5}{s+2}$$

$$\begin{aligned} y(t) &= -1t e^{-t} u(t) + 7e^{-t} u(t) - 5e^{-2t} u(t) \\ &= \underline{\underline{[-te^{-t} + 7e^{-t} - 5e^{-2t}]} u(t)} \end{aligned}$$

→ Find the impulse and step response of the following system.

$$(i) \quad H(s) = \frac{10}{s^2 + 6s + 10}$$

To find $y(t)$.

Impulse input $x(t) = \delta(t) \rightarrow x(s) = 1$

$$\frac{Y(s)}{X(s)} = H(s) = \frac{10}{s^2 + 6s + 10}$$

$$Y(s) = H(s) \cdot X(s) = \frac{10}{s^2 + 6s + 10} \times 1$$

$$Y(s) = \frac{10}{s^2 + 6s + 10}$$

$$= \frac{10}{(s+3)^2 + 1^2}$$

Taking Inv. LT

$$y(t) = 10 \cdot e^{-3t} \sin t \cdot u(t)$$

$$J = 6 \\ P = 10$$

$$s^2 + 6s + 10$$

$$s^2 + 6s + 9 + 1$$

$$(s^2 + 2 \cdot 3s + 3^2) + 1$$

$$(s+3)^2 + 1^2$$

$$L^{-1} \left[\frac{1}{s^2 + a^2} \right] = \sin at$$

$$L^{-1} \left[\frac{1}{(s+b)^2 + a^2} \right] = e^{-bt} \sin at$$

For a unit step input.

$$x(t) = u(t) \rightarrow X(s) = \frac{1}{s}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{10}{s^2 + 6s + 10}$$

$$Y(s) = H(s) \cdot X(s) = \frac{10}{s^2 + 6s + 10} \cdot \frac{1}{s}$$

$$= \frac{10}{s(s^2 + 6s + 10)}$$

$$Y(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 6s + 10}$$

$$A = \frac{10}{s(s^2+6s+10)} \cdot s \Big|_{s=0} = \frac{10}{10} = 1$$

$$\frac{10}{s(s^2+6s+10)} = \frac{A(s^2+6s+10) + (Bs+C)s}{s(s^2+6s+10)}$$

$$10 = s^2 + 6s + 10 + Bs^2 + Cs$$

$$10 = (B+1)s^2 + (6+C)s + 10$$

$$(B+1)s^2 + (6+C)s = 0$$

$$(B+1)s^2 = -(6+C)s$$

$$(B+1)s = -(6+C)$$

Eq. coefficients

$$(B+1) = 0 \rightarrow B = -1$$

$$(6+C) = 0 \rightarrow C = -6$$

$$\begin{aligned} Y(s) &= \frac{1}{s} + \frac{-s-6}{s^2+6s+10} \\ &= \frac{1}{s} - \frac{s+6}{s^2+6s+10} \end{aligned}$$

$$= \frac{1}{s} - \frac{s+6}{(s+3)^2+1^2}$$

$$= \frac{1}{s} - \frac{s+3}{(s+3)^2+1^2} - \frac{3}{(s+3)^2+1^2}$$

$$y(t) = \left[1 - e^{-3t} \cos t - 3 e^{-3t} \sin t \right] u(t)$$

$$= \underline{\underline{1 - e^{-3t} (\cos t + 3 \sin t) u(t)}}$$

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Concept of positive real function and Hurwitz polynomial

$$f(s) = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}$$

Necessary conditions for stability

1. Degree of numerator of $F(s)$ should not exceed the degree of denominator by more than unity. i.e., $(m-n) \leq 1$
2. $F(s)$ should not have multiple poles on $j\omega$ -axis
3. $F(s)$ should not have poles on the right half of the s -plane.

Hurwitz Polynomial

$P(s)$ is said to be Hurwitz polynomial if it satisfies the following basic properties

1. The polynomial $P(s)$ is real when 's' is real.
2. The roots of $P(s)$ have real parts which are either zero or negative.

Properties of Hurwitz Polynomial

1. All the coefficients are real and ^{non}negative i.e., all coefficients from a_n to a_0 are +ve.
2. All the terms starting from highest power of s to lowest power of s should be present. or no power of s should be missing in between, unless the polynomial is either even or odd.

- A polynomial is said to be even, if it consists of only even powers of s .

All odd powers of s are missing

$$P(s) = s^4 + 2s^2 + 8 \rightarrow \text{no odd powers should be missing}$$

- The polynomial is said to be odd, if it consists of only odd powers of s .

All even powers of s are missing

$$P(s) = s^5 + 2s^3 + 5s \rightarrow \text{no even powers should be missing}$$

→ 3. If the polynomial is even or odd, then all its roots must lie on imaginary plane

4. Hurwitz polynomial can be separated into even and odd parts.

$$P(s) = n(s) + m(s)$$

$n(s) \rightarrow$ odd part

$m(s) \rightarrow$ even part

If the ratio of odd to even parts of $P(s)$ or even to odd parts of $P(s)$ is expressed in continued fraction expansion, then all the quotient terms must be positive.

$$\psi(s) = \frac{m(s)}{n(s)} \quad \text{or} \quad \frac{n(s)}{m(s)}$$

$$\psi(s) = q_1s + \frac{1}{q_2s + \frac{1}{q_3s + \frac{1}{\dots + q_ns}}}$$

$q_1, q_2, q_3, \dots, q_n \rightarrow$ quotients must be positive for $P(s)$ to be Hurwitz.

→ Highest power polynomial out of $n(s)$ & $m(s)$ will be the numerator.

→ No. of quotients = No. of highest power of s

→ If $P(s)$ is even or odd, then it is not possible to express it in continued fraction form as either $n(s)$ or $m(s)$ is absent. So, it is done with its derivative $P'(s)$

$$P(s) = s^4 + 2s^2 + 3$$

$$P'(s) = 4s^3 + 4s$$

$\frac{P(s)}{P'(s)}$ → continued fraction expansion.

Test whether the polynomial

$$P(s) = s^3 + 4s^2 + 5s + 2 \text{ is Hurwitz.}$$

Even part of $P(s) \Rightarrow e(s) = 4s^2 + 2$

Odd part of $P(s) \Rightarrow o(s) = s^3 + 5s$
 (degree more in $o(s)$)

Continued fraction expansion of $\frac{o(s)}{e(s)}$.

$$\begin{array}{r} 4s^2 + 2 \overline{) s^3 + 5s} \left(\frac{s}{4} \right. \\ \underline{s^3 + \frac{5s}{4}} \\ \frac{9s}{4} \left(\frac{8s}{9} \right. \\ \underline{4s^2 + 2} \\ 2 \overline{) \frac{9s}{4}} \left(\frac{9s}{4} \right. \\ \underline{\frac{9s}{4}} \\ 0 \end{array}$$

$$5s - \frac{s}{2}$$

$$\frac{10s - s}{2} = \frac{9s}{2}$$

$$\frac{9s}{2} \times \frac{2 \times 4s - 4s^2}{9}$$

$$2 \times \frac{1 \times 9s}{2 \times 2}$$

All the quotient terms are +ve.
 $\therefore P(s)$ is Hurwitz.

→ Test whether the polynomial

$$P(s) = s^4 + s^3 + 3s^2 + 2s + 12 \text{ is Hurwitz.}$$

A: $e(s) = s^4 + 3s^2 + 12$

$$o(s) = s^3 + 2s$$

$$\begin{array}{r} s^3 + 2s \overline{) s^4 + 3s^2 + 12} \quad (s \\ \underline{s^4 + 2s^2} \\ s^2 + 12 \end{array} \quad \begin{array}{r} s^3 + 2s \overline{) s^3 + 12s} \\ \underline{s^3 + 12s} \\ -10s \end{array} \quad \begin{array}{r} s^2 + 12 \overline{) -10s} \quad \left(-\frac{1}{10}s \right. \\ \underline{-s^2 - 1.2} \\ 12.1 \end{array} \quad \begin{array}{r} -10s \overline{) 12.1} \quad \left(-\frac{10}{12}s \right. \\ \underline{-10s} \\ 0 \end{array}$$

All the quotient are not +ve. $\therefore P(s)$ is not Hurwitz

→ Test whether $P(s) = s^5 + 3s^3 + 2s$ is Hurwitz.

A: $P(s)$ is a odd polynomial. To find continued fraction expansion, derivative of $P(s)$ is needed

$$P'(s) = 5s^4 + 9s^2 + 2$$

$$\frac{P(s)}{P'(s)}$$

$$\begin{array}{r} s^4 + 9s^2 + 2 \quad) \quad s^5 + 3s^3 + 2s \quad \left(\frac{s}{5} \right. \\ \underline{s^5 + 9s^3 + 2s} \end{array}$$

$$\begin{array}{r} \frac{6}{5}s^3 + \frac{8}{5}s \quad) \quad s^4 + 9s^2 + 2 \quad \left(\frac{25s}{6} \right. \\ \underline{s^4 + 20s^2} \end{array}$$

$$\begin{array}{r} \frac{7s^2}{3} + 2 \quad) \quad \frac{6s^3}{5} + \frac{8s}{5} \quad \left(\frac{18s}{3} \right. \\ \underline{\frac{6s^3}{5} + \frac{36s}{35}} \end{array}$$

$$\begin{array}{r} \frac{20}{35}s \quad) \quad \frac{7s^2}{3} + 2 \quad \left(\frac{49}{12}s \right. \\ \underline{\frac{7s^2}{3}} \end{array}$$

$$\begin{array}{r} 2 \quad) \quad \frac{20}{35}s \quad \left(\frac{10}{35}s \right. \\ \underline{\frac{20}{35}s} \end{array}$$

$$\begin{array}{r} \frac{20}{35}s \\ \underline{} \\ 0 \end{array}$$

$$3 - \frac{9}{5} = \frac{15-9}{5} = \frac{6}{5}$$

$$2 - \frac{2}{5} = \frac{10-2}{5} = \frac{8}{5}$$

$$\frac{6s^3}{5} \times \frac{5s}{6}$$

$$\frac{525}{6} \times \frac{4}{5}$$

$$9 - \frac{20}{3} = \frac{27-20}{3} = \frac{7}{3}$$

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All quotients are positive, $\therefore P(s)$ is Hurwitz.

Positive Real function

$F(s) = \frac{A(s)}{B(s)}$ is positive real function if

it fulfills the following conditions.

1. Both $A(s)$ and $B(s)$ should be Hurwitz polynomial. Poles and zeros of $F(s)$ should lie on left half of s -plane.
2. Highest and lowest powers of $A(s)$ & $B(s)$ differ by at most 1. (max. diff $m-n=1$)
3. If $F(s)$ is PR fn, then $1/F(s)$ is also PRF.
4. The sum of PRF is also a PR function. (but the diff. of two PRF is not necessarily a PR fn).

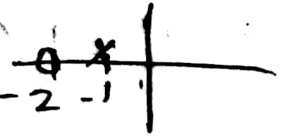
$$F(s) = \frac{s+2}{s+1}$$

Test to check PRF.

1. Check all coeff. of Nr & Dr are +ve.
2. Check poles & zeros lie on left half of s-plane. If poles on imaginary axis, poles are simple and with ^{neg} positive residues.
3. Sub $s = j\omega$ in $F(s)$.
4. $\text{Re}[F(j\omega)] \geq 0$.

$$\rightarrow F(s) = \frac{s+2}{s+1}$$

1. All coeff are +ve.
2. Pole = -1, Zero = -2.



All roots lie on left half of s-plane.

$$\begin{aligned}
 3. \quad F(s) &= \frac{j\omega+2}{j\omega+1} \cdot \frac{j\omega-1}{j\omega-1} = \frac{(j\omega)^2 - j\omega + 2j\omega - 2}{(j\omega)^2 - 1^2} \\
 &= \frac{-\omega^2 + j\omega - 2}{-\omega^2 - 1}
 \end{aligned}$$

$$= \frac{\omega^2 + j\omega + 2}{\omega^2 + 1}$$

$$= \frac{\omega^2 + 2}{\omega^2 + 1} - \frac{j\omega}{\omega^2 + 1}$$

$$4. \operatorname{Re}[F(j\omega)] = \frac{\omega^2 + 2}{\omega^2 + 1} \geq 0 \text{ for all } \omega.$$

$\therefore F(s)$ is a positive real function.

$$\rightarrow F(s) = \frac{s^3 + 6s^2 + 7s + 3}{s^2 + 2s + 1}$$

1. All coeff. are +ve.

2. To check roots lie on left half of s-plane Routh Hurwitz criterion can be done.

$$N(s) = s^3 + 6s^2 + 7s + 3$$

$$D(s) = s^2 + 2s + 1$$

s^3	1	7
s^2	6	3
s^1	$\frac{42-3}{6} = \frac{39}{6} = 6.5$	
s^0	3	

s^2	1	1
s	2	0
s^0	1	

All +ve in 1st column. $\therefore$ all roots lie on left half

$$\begin{aligned}
 3. \quad F(j\omega) &= \frac{(j\omega)^3 + 6(j\omega)^2 + 7j\omega + 3}{(j\omega)^2 + 2j\omega + 1} \\
 &= \frac{-j\omega^3 - 6\omega^2 + 7j\omega + 3}{-\omega^2 + 2j\omega + 1} \\
 &= \frac{+j\omega^3 + 6\omega^2 - 7j\omega - 3}{\omega^2 - 2j\omega + 1} \times \frac{\omega^2 + 1 + 2j\omega}{\omega^2 + 1 + 2j\omega} \\
 &= \frac{j\omega^5 + 6\omega^4 - 7j\omega^3 - 3\omega^2 + j\omega^3 + 6\omega^2 - 7j\omega - 3}{(\omega^2 + 1)^2 - (2j\omega)^2} \\
 &= \frac{j\omega^5 + 6\omega^4 - 6j\omega^3 + 3\omega^2 - 7j\omega - 3}{\omega^4 + 4\omega^2 + 1 - 4\omega^2} \\
 &= \frac{j\omega^5 + 6\omega^4 - 6j\omega^3 + 3\omega^2 - 7j\omega - 3}{\omega^4 + 1}
 \end{aligned}$$

Even $M_1(s) = 6s^2 + 3$

$M_2(s) = s^2 + 1$

Odd $N_1(s) = s^3 + 7s$

$N_2(s) = 2s$

$$M_1 M_2 - N_1 N_2 = (6s^2 + 3)(s^2 + 1) - (s^3 + 7s)2s \Big|_{s=j\omega}$$

$$= 6s^4 + 6s^2 + 3s^2 + 3 - 2s^4 - 14s^2$$

$$= 4s^4 - 5s^2 + 3 \Big|_{s=j\omega}$$

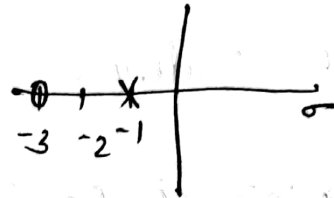
$$= 4\omega^4 + 5\omega^2 + 3 > 0, \therefore F(s) \text{ is PR}$$

→ Test whether $F(s)$ is a positive real function.

$$F(s) = \frac{s+3}{s+1}$$

A: Pole at $s = -1$

Zero at $s = -3$.



Pole & Zero are at left half of s-plane

$$F(s) = \frac{N(s)}{D(s)} = \frac{s+3}{s+1} \quad \frac{j\omega+3}{j\omega+1} \times \frac{j\omega-1}{j\omega-1}$$

Even part of $N(s), m_1 = 3$

Odd part of $N(s), n_1 = s$

Even part of $D(s), m_2 = 1$

Odd part of $D(s), n_2 = s$

$$\frac{-\omega^2 - j\omega + 3j\omega - 3}{(j\omega)^2 - 1}$$

$$\frac{-\omega^2 + 2j\omega - 3}{-\omega^2 - 1}$$

$$\frac{\omega^2 - 2j\omega + 3}{\omega^2 + 1}$$

$$\frac{\omega^2 + 3}{\omega^2 + 1} - j \frac{2\omega}{\omega^2 + 1}$$

$$R[F(j\omega)] \geq 0$$

$$A(\omega^2) = m_1 m_2 - n_1 n_2 = 3 - s^2 \Big|_{s=j\omega}$$

$$= 3 - j^2 \omega^2$$

$$= 3 + \omega^2$$

$A(\omega^2)$ is +ve for all values of ω . $\therefore F(s)$ is PR

Signal Flow Graph - Mason's Gain formula

→ A signal flow graph is a diagram that represents a set of simultaneous linear algebraic equations. By taking LT, the time domain diff. eq governing a control system can be transferred to a set of algebraic equations in s-domain. The signal flow graph of the system can be constructed using these equations.

→ A signal flow graph consists of a network in which nodes are connected by directed branches.

Node → represent a system variable.

Branch → connected between nodes acts as a signal multiplier.

→ Signal flows only in one direction.

Direction of signal flow is indicated by an arrow placed on the branch and gain is indicated along the branch.

Node → Represents variable or signal.

Branch → is directed line segment joining 2 nodes.

~~with~~ arrow → direction, gain → transmittance

Transmittance — gain acquired by the signal when it travels from one node to another.

Input node (source) — Node that has only outgoing branches.

Output node (Sink) — Node that has only incoming branches.

Mixed node — Node that has both incoming and outgoing branches.

Path $\rightarrow$ Traversal of connected branches in the direction of branch arrows. The path should not cross a node more than once.

Open path $\rightarrow$ A open path starts at a node and ends at another node.

Closed path $\rightarrow$ Starts and ends at same node.

Forward path $\rightarrow$ Path from an input node to an output node that does not cross any node more than once.

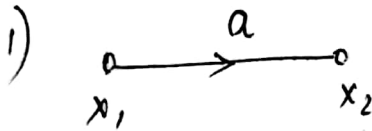
Forward path gain $\rightarrow$ Product of branch transmittances (gain) of a forward path.

Individual loop $\rightarrow$ Closed path starting from a node and after passing through a certain part of a graph arrives at same node without crossing any node more than once.

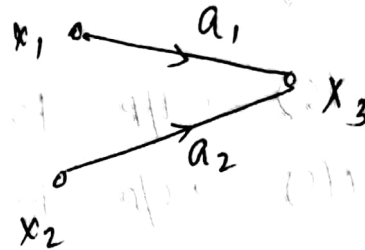
Loop gain $\rightarrow$ Product of branch transmittances (gain) of a loop.

Non-touching loop - If the loops doesnot have a common node, then they are said to be a non-touching loops.

Rules

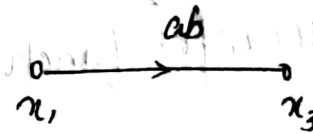
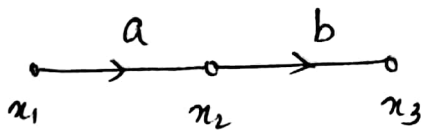


$$x_2 = a x_1$$

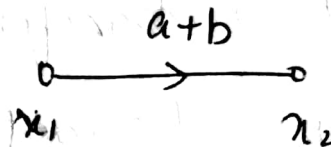
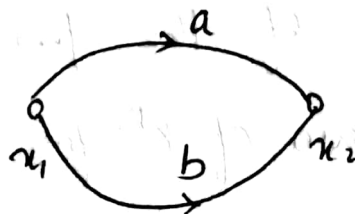


$$x_3 = a_1 x_1 + a_2 x_2$$

2) Cascaded branches



3) Parallel branches



Signal flow graph reduction

Mason's gain formula

used to determine the transfer function of the system from the signal flow graph.

Let $R(s) \rightarrow$ i/p to the s/m.

$C(s) \rightarrow$ o/p to the s/m.

Transfer function of the s/m, $T(s) = \frac{C(s)}{R(s)}$

Mason's gain formula states that the overall gain [transfer function]

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

$T \rightarrow$ transfer function of the s/m.

$P_k \rightarrow$ forward path gain of k^{th} forward path.

$k \rightarrow$ No. of forward path in ~~k^{th}~~ signal flow graph.

$$\Delta = 1 - (\text{sum of individual loop gains})$$

$$+ \left\{ \text{sum of gain products of all possible combinations of two non-touching loops} \right\}$$

$$- \left(\text{sum of gain products of all possible combinations of three non-touching loops} \right)$$

$$+ \dots$$

$\Delta_k = \Delta$ for that part of the graph which is not touching k^{th} forward path

PROCEDURE FOR CONVERTING BLOCK DIAGRAM TO SIGNAL FLOW GRAPH

The signal flow graph and block diagram of a system provides the same information but there is no standard procedure for reducing the block diagram to find the transfer function of the system. Also the block diagram reduction technique will be tedious and it is difficult to choose the rule to be applied for simplification. Hence it will be easier if the block diagram is converted to signal flow graph and **Mason's gain formula** is applied to find the transfer function. The following procedure can be used to convert block diagram to signal flow graph.

1. Assume nodes at input, output, at every summing point, at every branch point and in between cascaded blocks.
2. Draw the nodes separately as small circles and number the circles in the order 1, 2, 3, 4, etc.
3. From the block diagram find the gain between each node in the main forward path and connect all the corresponding circles by straight line and mark the gain between the nodes.
4. Draw the feed forward paths between various nodes and mark the gain of feed forward path along with sign.
5. Draw the feedback paths between various nodes and mark the gain of feedback paths along with sign.

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EXAMPLE 1.25

Find the overall transfer function of the system whose signal flow graph is shown in fig 1.

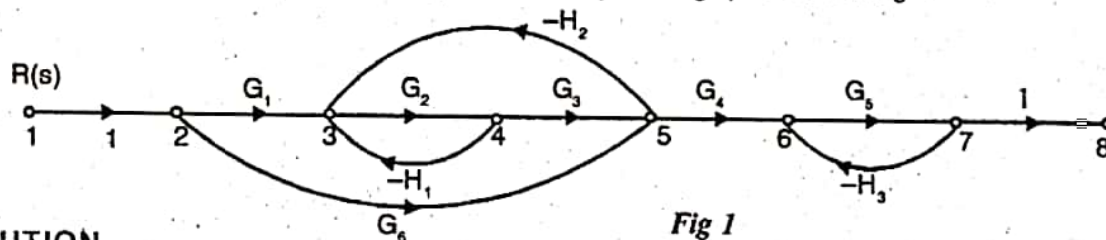


Fig 1

SOLUTION**Forward Path Gains**

There are two forward paths. $\therefore K = 2$

Let forward path gains be P_1 and P_2 .

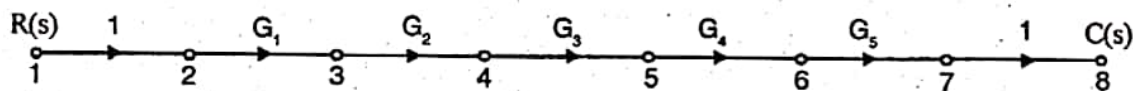


Fig 2 : Forward path-1.

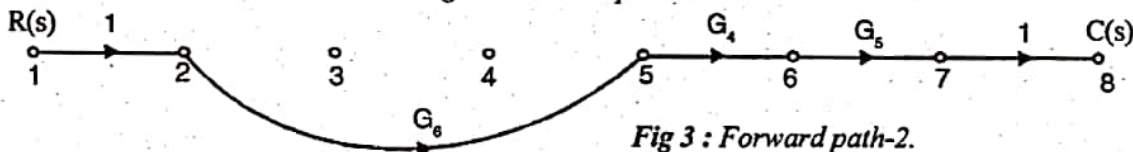


Fig 3 : Forward path-2.

Gain of forward path-1, $P_1 = G_1 G_2 G_3 G_4 G_5$

Gain of forward path-2, $P_2 = G_6 G_4 G_5$

Individual Loop Gain

There are three individual loops. Let individual loop gains be P_{11} , P_{21} and P_{31} .

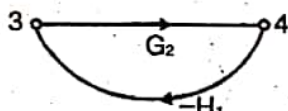


Fig 4 : Loop-1.

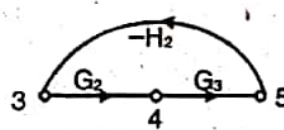


Fig 5 : Loop-2.

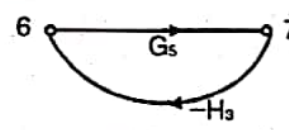


Fig 6 : Loop-3.

Loop gain of individual loop-1, $P_{11} = -G_2 H_1$

Loop gain of individual loop-2, $P_{21} = -G_2 G_3 H_2$

Loop gain of individual loop-3, $P_{31} = -G_5 H_3$

1. Gain Products of Two Non-touching Loops

There are two combinations of two non-touching loops. Let the gain products of two non touching loops be P_{12} and P_{22} .

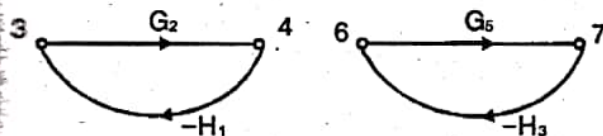


Fig 7 : First combination of 2 non-touching loops.

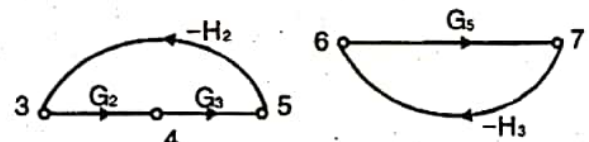


Fig 8 : Second combination of 2 non-touching loops.

Gain product of first combination of two non touching loops $\left. \begin{array}{l} \\ \end{array} \right\} P_{12} = P_{11} P_{31} = (-G_2 H_1) (-G_5 H_3) = G_2 G_5 H_1 H_3$

Gain product of second combination of two non touching loops $\left. \begin{array}{l} \\ \end{array} \right\} P_{22} = P_{21} P_{31} = (-G_2 G_3 H_2) (-G_5 H_3) = G_2 G_3 G_5 H_2 H_3$

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IV. Calculation of Δ and Δ_K

$$\begin{aligned}\Delta &= 1 - (P_{11} + P_{21} + P_{31}) + (P_{12} + P_{22}) \\ &= 1 - (-G_2H_1 - G_2G_3H_2 - G_5H_3) + (G_2G_5H_1H_3 + G_2G_3G_5H_2H_3) \\ &= 1 + G_2H_1 + G_2G_3H_2 + G_5H_3 + G_2G_5H_1H_3 + G_2G_3G_5H_2H_3\end{aligned}$$

$\Delta_1 = 1$, Since there is no part of graph which is not touching with first forward path.

The part of the graph which is non touching with second forward path is shown in fig 9.

$$\Delta_2 = 1 - P_{11} = 1 - (-G_2H_1) = 1 + G_2H_1$$

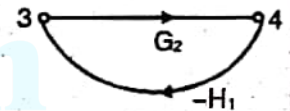


Fig 9

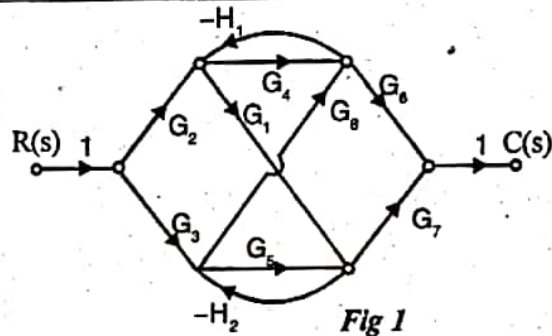
V. Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$\begin{aligned}T &= \frac{1}{\Delta} \sum_K P_K \Delta_K = \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2) \quad (\text{Number of forward paths is 2 and so } K = 2) \\ &= \frac{G_1G_2G_3G_4G_5 + G_4G_5G_6 (1 + G_2H_1)}{1 + G_2H_1 + G_2G_3H_2 + G_5H_3 + G_2G_5H_1H_3 + G_2G_3G_5H_2H_3} \\ &= \frac{G_1G_2G_3G_4G_5 + G_4G_5G_6 + G_2G_4G_5G_6H_1}{1 + G_2H_1 + G_2G_3H_2 + G_5H_3 + G_2G_5H_1H_3 + G_2G_3G_5H_2H_3} \\ &= \frac{G_2G_4G_5 [G_1G_3 + G_6 / G_2 + G_6H_1]}{1 + G_2H_1 + G_2G_3H_2 + G_5H_3 + G_2G_5H_1H_3 + G_2G_3G_5H_2H_3}\end{aligned}$$

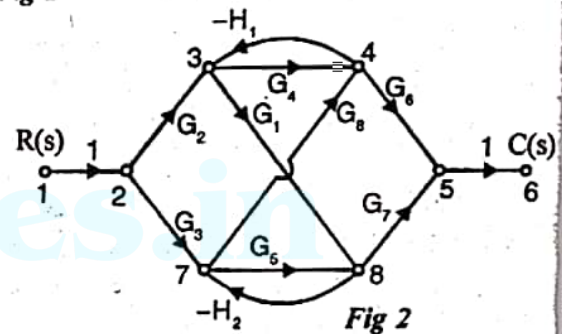
EXAMPLE 1.26

Find the overall gain of the system whose signal flow graph is shown in fig 1.



SOLUTION

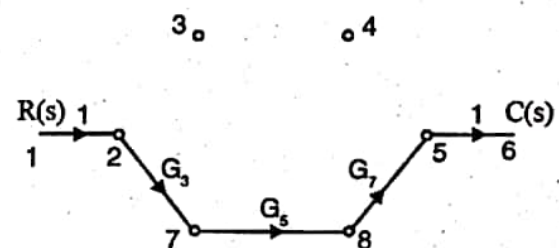
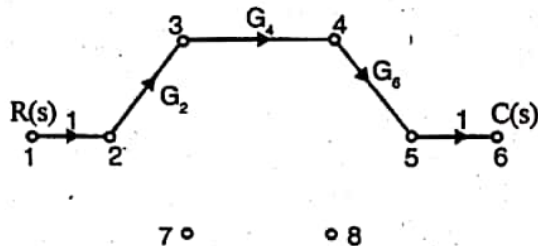
Let us number the nodes as shown in fig 2.



I. Forward Path Gains

There are six forward paths. $\therefore K = 6$

Let the forward path gains be P_1, P_2, P_3, P_4, P_5 and P_6 .



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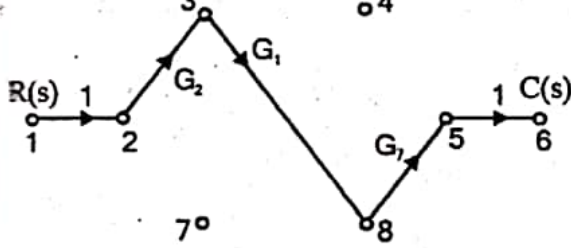


Fig 5 : Forward path-3

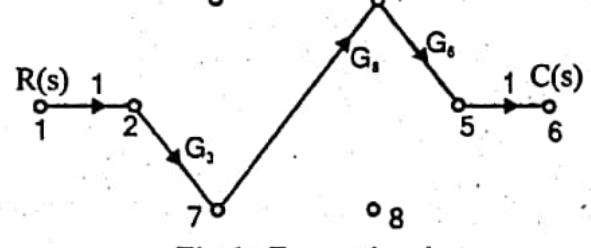


Fig 6 : Forward path-4

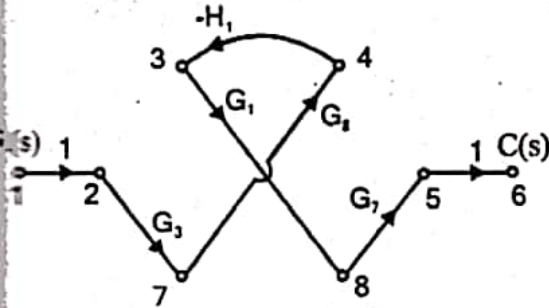


Fig 7 : Forward path-5

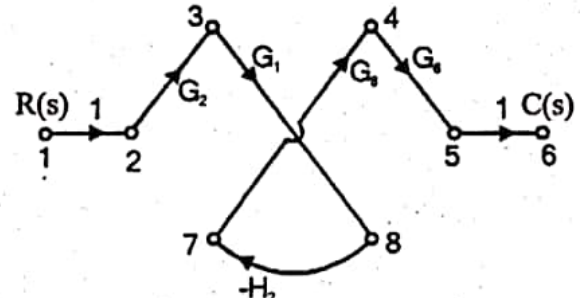


Fig 8 : Forward path-6

Gain of forward path-1, $P_1 = G_2 G_4 G_6$

Gain of forward path-2, $P_2 = G_3 G_5 G_7$

Gain of forward path-3, $P_3 = G_1 G_2 G_7$

Gain of forward path-4, $P_4 = G_3 G_8 G_6$

Gain of forward path-5, $P_5 = -G_1 G_3 G_7 G_8 H_1$

Gain of forward path-6, $P_6 = -G_1 G_2 G_6 G_8 H_2$

Individual Loop Gain

There are three individual loops.

Let individual loop gains be P_{11} , P_{21} and P_{31} .

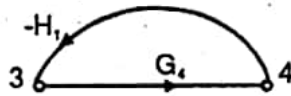


Fig 9 : Loop-1

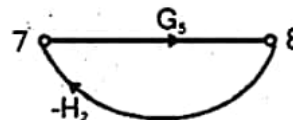


Fig 10 : Loop-2

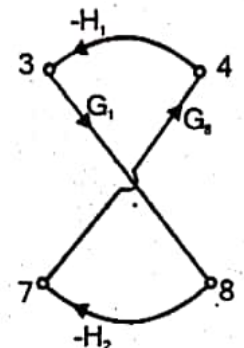


Fig 11 : Loop-3

Loop gain of individual loop-1, $P_{11} = -G_4 H_1$

Loop gain of individual loop-2, $P_{21} = -G_5 H_2$

Loop gain of individual loop-3, $P_{31} = G_1 G_8 H_1 H_2$

Gain Products of Two Non-touching Loops

There is only one combination of two non-touching

loops. Let gain product of two non-touching loops be P_{12} .

Gain product of first combination of two non-touching loops $\left. \begin{array}{l} \text{Gain product of first combination} \\ \text{of two non-touching loops} \end{array} \right\} P_{12} = P_{11} P_{21} = (-G_4 H_1) (-G_5 H_2) = G_4 G_5 H_1 H_2$

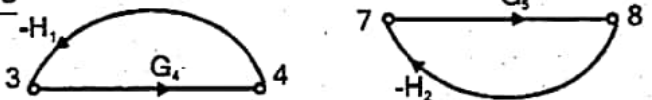


Fig 12 : Combination of 2 non-touching loops

Calculation of Δ and Δ_K

$$\Delta = 1 - (P_{11} + P_{21} + P_{31}) + P_{12} = 1 - (-G_4 H_1 - G_5 H_2 + G_1 G_8 H_1 H_2) + G_4 G_5 H_1 H_2$$

$$= 1 + G_4 H_1 + G_5 H_2 - G_1 G_8 H_1 H_2 + G_4 G_5 H_1 H_2$$

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The part of the graph non-touching forward path - 1 is shown in fig 13.

$$\therefore \Delta_1 = 1 - (-G_5 H_2) = 1 + G_5 H_2$$

The part of the graph non-touching forward path -2 is shown in fig 14.

$$\therefore \Delta_2 = 1 - (-G_4 H_1) = 1 + G_4 H_1$$

There is no part of the graph which is non-touching with forward paths 3, 4, 5 and 6.

$$\therefore \Delta_3 = \Delta_4 = \Delta_5 = \Delta_6 = 1$$

V. Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$\begin{aligned} T &= \frac{1}{\Delta} \left(\sum_K P_K \Delta_K \right) \quad (\text{Number of forward paths is six and so } K = 6) \\ &= \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2 + P_3 \Delta_3 + P_4 \Delta_4 + P_5 \Delta_5 + P_6 \Delta_6) \\ &= \frac{G_2 G_4 G_6 (1 + G_5 H_2) + G_3 G_5 G_7 (1 + G_4 H_1) + G_1 G_2 G_7 + G_3 G_6 G_8}{1 + G_4 H_1 + G_5 H_2 - G_1 G_3 G_7 G_8 H_1 - G_1 G_2 G_6 G_8 H_2 + G_4 G_5 H_1 H_2} \end{aligned}$$

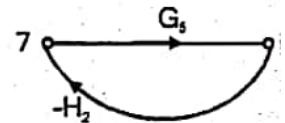


Fig 13

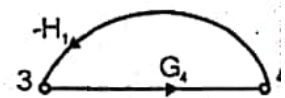


Fig 14

EXAMPLE 1.28

Find the overall gain $C(s)/R(s)$ for the signal flow graph shown in fig 1.

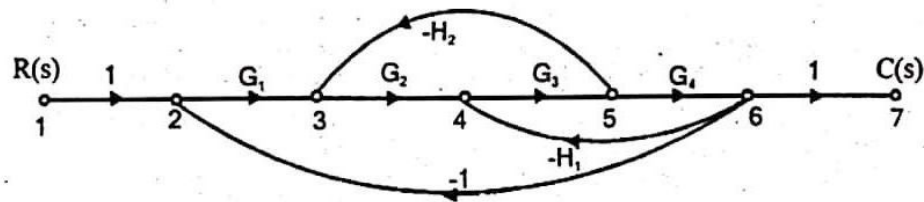


Fig 1

SOLUTION**I. Forward Path Gains**

There is only one forward path. $\therefore K = 1$.

Let the forward path gain be P_1 .

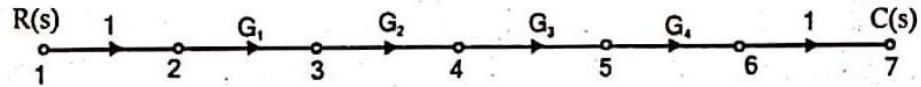


Fig 1 : Forward path-1

Gain of forward path-1, $P_1 = G_1 G_2 G_3 G_4$

II. Individual Loop Gain

There are three individual loops. Let the loop gains be P_{11}, P_{21}, P_{31} .

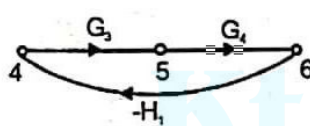


Fig 3 : loop-1

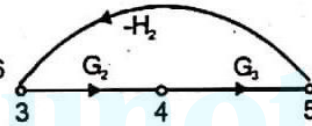


Fig 4 : loop-2

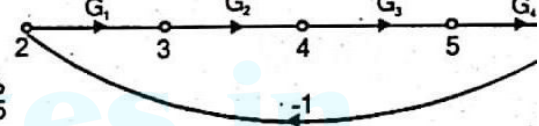


Fig 5 : loop-3

Loop gain of individual loop-1, $P_{11} = -G_3 G_4 H_1$

Loop gain of individual loop-2, $P_{21} = -G_2 G_3 H_2$

Loop gain of individual loop-3, $P_{31} = -G_1 G_2 G_3 G_4$

III. Gain Products of Two Non-touching Loops

There are no possible combinations of two non-touching loops, three non-touching loops, etc.

IV. Calculation of Δ and Δ_K

$$\begin{aligned}\Delta &= 1 - (P_{11} + P_{21} + P_{31}) \\ &= 1 - (-G_3 G_4 H_1 - G_2 G_3 H_2 - G_1 G_2 G_3 G_4) \\ &= 1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4\end{aligned}$$

Since no part of the graph is non-touching with forward path-1, $\Delta_1 = 1$.

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Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$\begin{aligned}T &= \frac{C(s)}{R(s)} = \frac{1}{\Delta} \sum_K P_K \Delta_K = \frac{1}{\Delta} P_1 \Delta_1 \quad (\text{Number of forward path is 1 and so } K = 1) \\ &= \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4}\end{aligned}$$

EXAMPLE 1.30

Convert the given block diagram to signal flow graph and determine $C(s)/R(s)$.

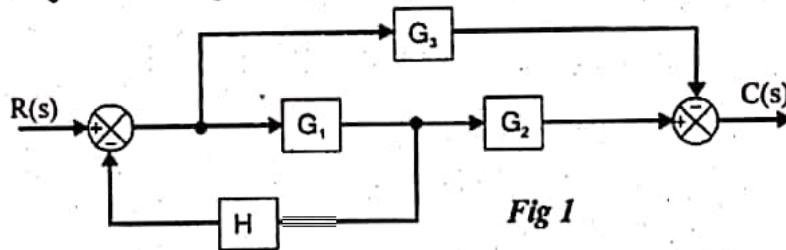


Fig 1

SOLUTION

The nodes are assigned at input, output, at every summing point & branch point as shown in fig 2.

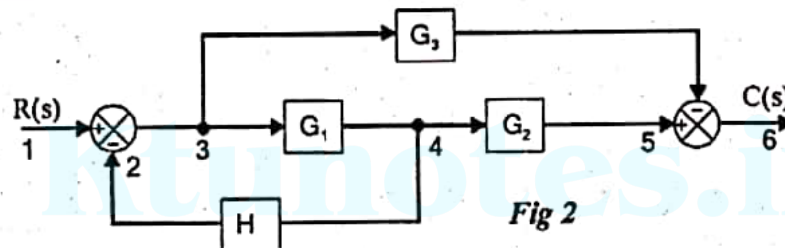


Fig 2

The signal flow graph of the above system is shown in fig 3.

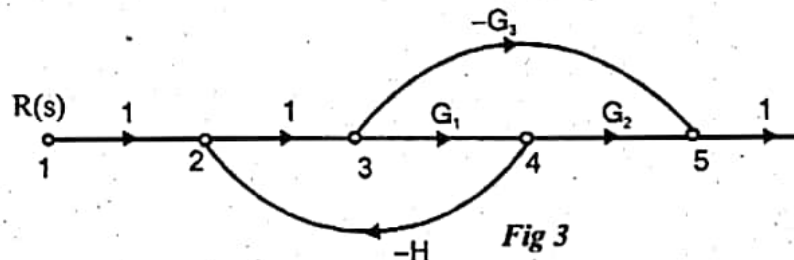


Fig 3

I. Forward Path Gains

There are two forward paths. $\therefore K=2$

Let the forward path gains be P_1 and P_2 .

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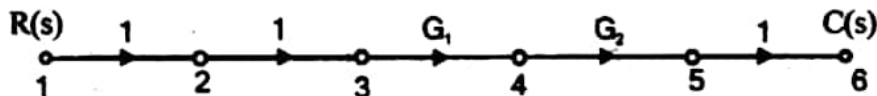


Fig 4 : Forward path-1



Fig 5 : Forward path-2

Gain of forward path-1, $P_1 = G_1 G_2$

Gain of forward path-2, $P_2 = -G_3$

Individual Loop Gain

There is only one individual loop. Let the individual loop gain be P_{11} .

Loop gain of individual loop-1, $P_{11} = -G_1 H$.

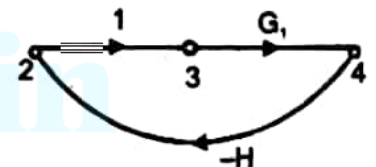


Fig 3 : loop-1

Gain Products of Two Non-touching Loops

There are no combinations of non-touching Loops.

Calculation of Δ and Δ_k

$$\Delta = 1 - [P_{11}] = 1 + G_1 H$$

Since there are no part of the graph which is non-touching with forward path-1 and 2,

$$\Delta_1 = \Delta_2 = 1$$

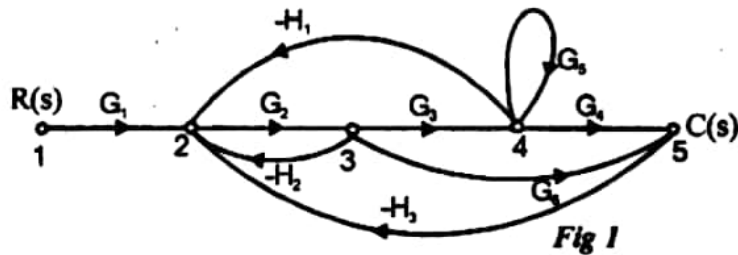
Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k = \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2] = \frac{G_1 G_2 - G_3}{1 + G_1 H}$$

EXAMPLE 1.27

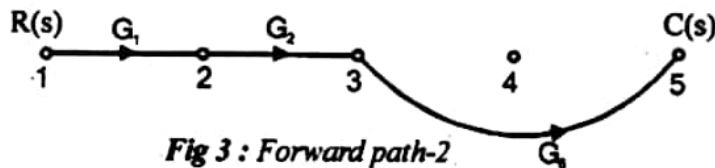
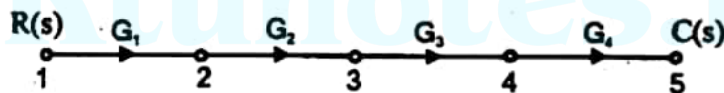
Find the overall gain $C(s)/R(s)$ for the signal flow graph shown in fig 1.



SOLUTION

I. Forward Path Gains

There are two forward paths. $\therefore K = 2$. Let the forward path gains be P_1 and P_2 .



Gain of forward path-1, $P_1 = G_1 G_2 G_3 G_4$

Gain of forward path-2, $P_2 = G_1 G_2 G_4$

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There are five individual loops. Let the individual loop gains be P_{11} , P_{21} , P_{31} , P_{41} and P_{51} .

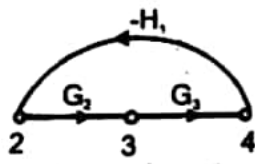


Fig 4 : loop-1

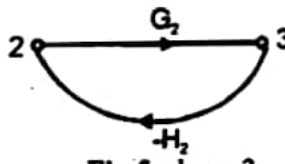


Fig 5 : loop-2

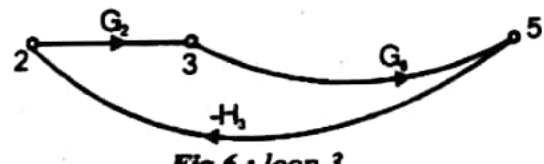


Fig 6 : loop-3

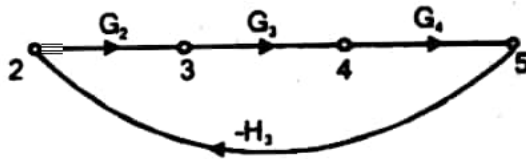


Fig 7 : loop-4



Fig 8 : loop-5

Loop gain of individual loop-1, $P_{11} = -G_2G_3H_1$

Loop gain of individual loop-2, $P_{21} = -H_2G_2$

Loop gain of individual loop-3, $P_{31} = -G_2G_5H_3$

Loop gain of individual loop-4, $P_{41} = -G_2G_3G_4H_4$

Loop gain of individual loop-5, $P_{51} = G_5$

Gain Products of Two Non-touching Loops

There are two combinations of two non-touching loops.

Let the gain products of two non-touching loops be P_{12} and P_{22} .

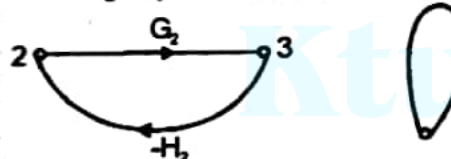


Fig 9 : First combination of two non-touching loops

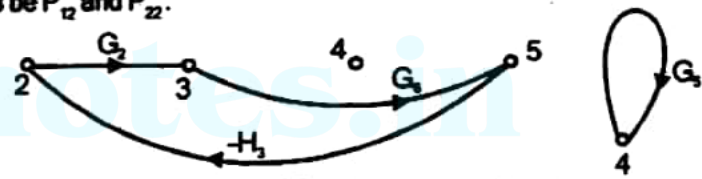


Fig 10 : Second combination of two non-touching loops

Gain product of first combination of two non touching loops $P_{12} = P_{21}P_{51} = (-G_2H_2)(G_5) = G_2G_5H_2$

Gain product of second combination of two non touching loops $P_{22} = P_{31}P_{51} = (-G_2G_5H_3)(G_5) = -G_2G_5G_5H_3$

Calculation of Δ and Δ_k

$$\begin{aligned}\Delta &= 1 - (P_{11} + P_{21} + P_{31} + P_{41} + P_{51}) + (P_{12} + P_{22}) \\ &= 1 - (-G_2G_3H_1 - H_2G_2 - G_2G_3G_4H_3 + G_5 - G_2G_5H_3) \\ &\quad + (-G_2H_2G_5 - G_2G_5G_5H_3)\end{aligned}$$

Since there is no part of graph which is not touching forward path-1, $\Delta_1 = 1$.

The part of graph which is not touching forward path-2 is shown in fig 11.

$$\therefore \Delta_2 = 1 - G_5$$

Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k \quad (\text{Number of forward path is 2 and each is } P_k)$$

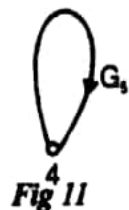
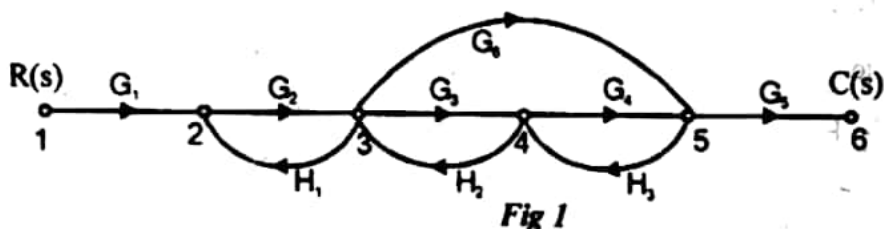


Fig 11

$$\begin{aligned}
 &= \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2] = \frac{1}{\Delta} [G_1 G_2 G_3 G_4 \times 1 + G_1 G_2 G_6 (1 - G_5)] \\
 &= \frac{G_1 G_2 G_3 G_4 + G_1 G_2 G_6 - G_1 G_2 G_5 G_6}{1 + G_2 G_3 H_1 + H_2 G_2 + G_2 G_3 G_4 H_3 - G_5 + G_2 G_6 H_3 - G_2 H_2 G_3 - G_2 G_3 G_4 H_3}
 \end{aligned}$$

KAMPLE 1.29

The signal flow graph for a feedback control system is shown in fig 1. Determine the closed loop transfer function $C(s)/R(s)$.

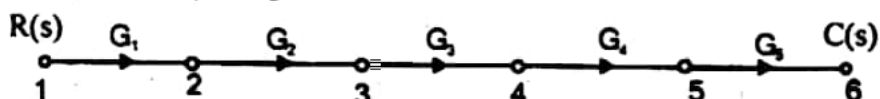


SOLUTION

Forward Path Gains

There are two forward paths. $\therefore K = 2$.

Let forward path gains be P_1 and P_2 .

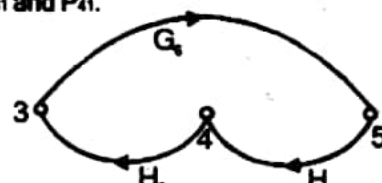
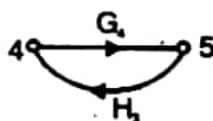
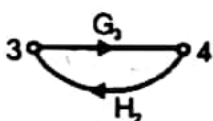
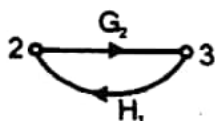


Gain of forward path-1, $P_1 = G_1 G_2 G_3 G_4 G_5$

Gain of forward path-2, $P_2 = G_1 G_2 G_6 G_5$

Individual Loop Gain

There are four individual loops. Let individual loop gains be P_{11} , P_{21} , P_{31} and P_{41} .



Loop gain of individual loop-1, $P_{11} = G_2 H_1$

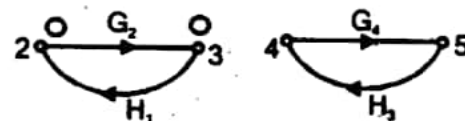
Loop gain of individual loop-2, $P_{21} = G_3 H_2$

Loop gain of individual loop-3, $P_{31} = G_4 H_3$

Loop gain of individual loop-4, $P_{41} = G_6 H_2 H_3$

Gain Products of Two Non-touching Loops

There is only one combination of two non-touching loops. Let the gain products of two non-touching loops be P_{12} .



$$\left. \begin{array}{l} \text{Gain product of first combination} \\ \text{of two non-touching loops} \end{array} \right\} P_{12} = (G_2 H_1) (G_4 H_3) \\ = G_2 G_4 H_1 H_3$$

IV. Calculation of Δ and Δ_K

$$\begin{aligned} \Delta &= 1 - (P_{11} + P_{21} + P_{31} + P_{41}) + P_{12} \\ &= 1 - (G_2 H_1 + G_3 H_2 + G_4 H_3 + G_6 H_2 H_3) + G_2 G_4 H_1 H_3 \\ &= 1 - G_2 H_1 - G_3 H_2 - G_4 H_3 - G_6 H_2 H_3 + G_2 G_4 H_1 H_3 \end{aligned}$$

Since there is no part of graph which is non-touching with forward path-1 and 2, $\Delta_1 = \Delta_2 = 1$

V. Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$\begin{aligned} T &= \frac{1}{\Delta} \sum_K P_K \Delta_K = \frac{1}{\Delta} (P_1 \Delta_1 + P_2 \Delta_2) \quad (\text{Number of forward paths is two and so } K = 2) \\ &= \frac{G_1 G_2 G_3 G_4 G_5 + G_1 G_2 G_5 G_6}{1 - G_2 H_1 - G_3 H_2 - G_4 H_3 - G_6 H_2 H_3 + G_2 G_4 H_1 H_3} \end{aligned}$$

EXAMPLE 1.31

Convert the block diagram to signal flow graph and determine the transfer function using Mason's gain formula.

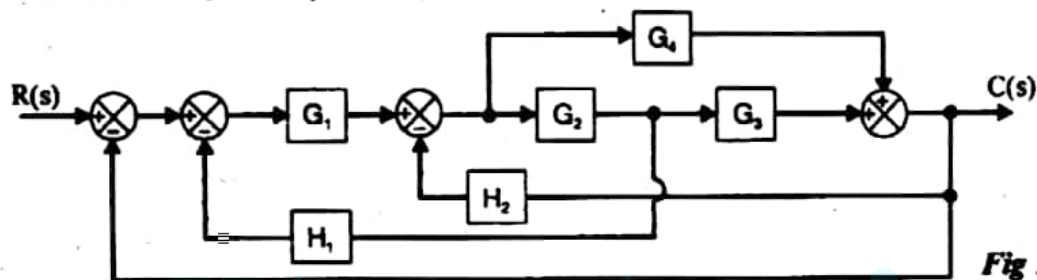
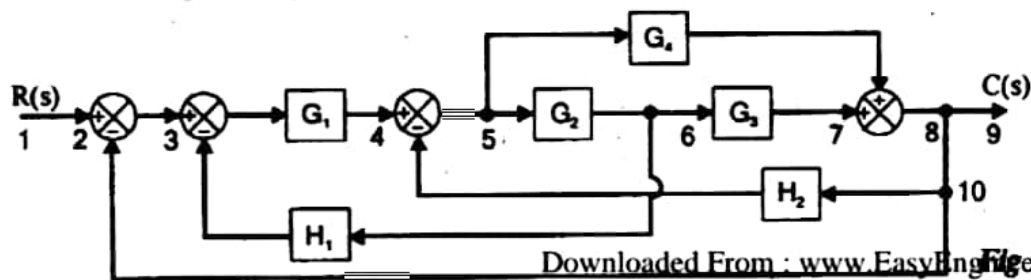


Fig 1

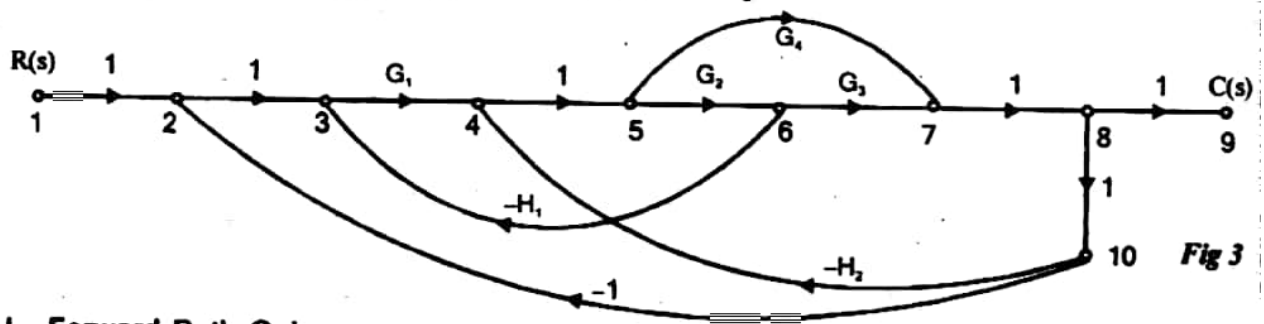
SOLUTION

The nodes are assigned at input, output, at every summing point & branch point as shown in fig 2.



Downloaded From : www.EasyEngineering.net Fig 2

The signal flow graph for the above block diagram is shown in fig 3.



I. Forward Path Gains

There are two forward paths. $\therefore K=2$.

Let the gain of the forward paths be P_1 and P_2 .

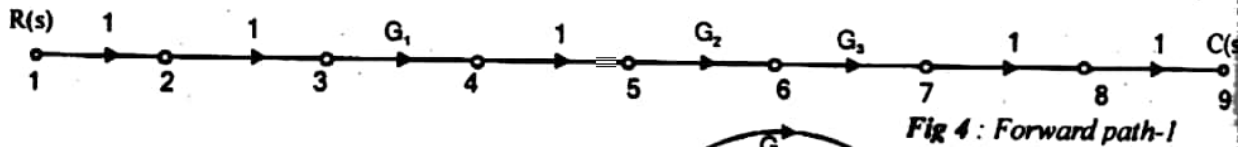


Fig 4 : Forward path-1

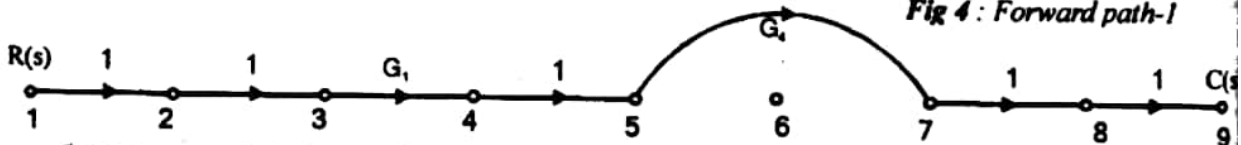


Fig 5 : Forward path-2

Gain of forward path-1, $P_1 = G_1G_2G_3$

Gain of forward path-2, $P_2 = G_1G_4$

II. Individual Loop Gain

There are five individual loops. Let the individual loop gain be $P_{11}, P_{21}, P_{31}, P_{41}$ and P_{51} .

Loop gain of individual loop-1, $P_{11} = -G_1G_2G_3$

Loop gain of individual loop-2, $P_{21} = -G_2G_1H_1$

Loop gain of individual loop-3, $P_{31} = -G_2G_3H_2$

Loop gain of individual loop-4, $P_{41} = -G_1G_4$

Loop gain of individual loop-5, $P_{51} = -G_4H_2$

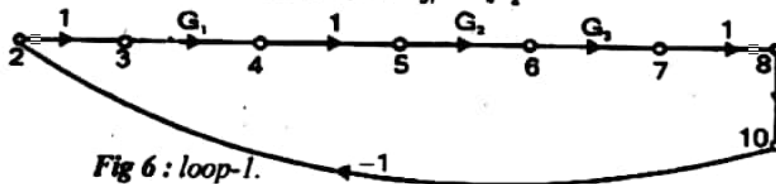


Fig 6 : loop-1.

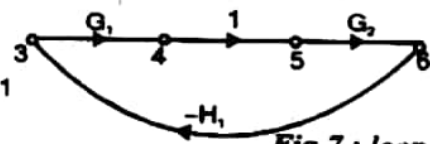


Fig 7 : loop-2.

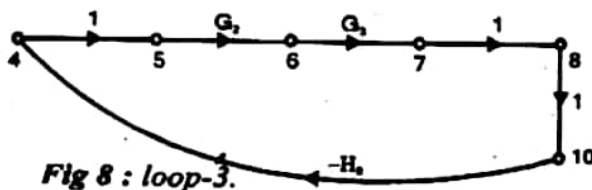


Fig 8 : loop-3.

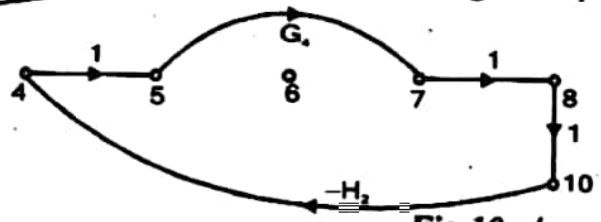


Fig 10 : loop-5.

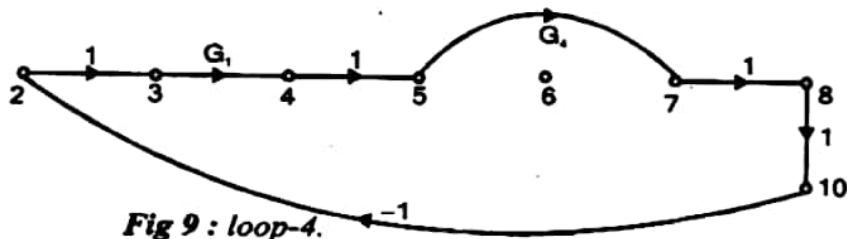


Fig 9 : loop-4.

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Gain Products of Two Non-touching Loops

There are no possible combinations of two non-touching loops, three non-touching loops, etc.,.

Calculation of Δ and Δ_k

$$\Delta = 1 - [P_{11} + P_{21} + P_{31} + P_{41} + P_{51}] = 1 + G_1 G_2 G_3 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_4 + G_4 H_2$$

Since no part of graph is non touching with forward paths-1 and 2, $\Delta_1 = \Delta_2 = 1$.

Transfer Function, T

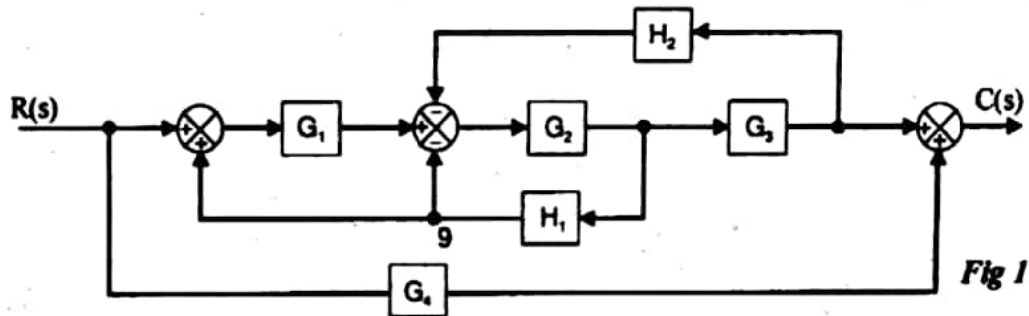
By Mason's gain formula the transfer function, T is given by,

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k = \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2]$$

$$= \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_1 G_2 G_3 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_4 + G_4 H_2}$$

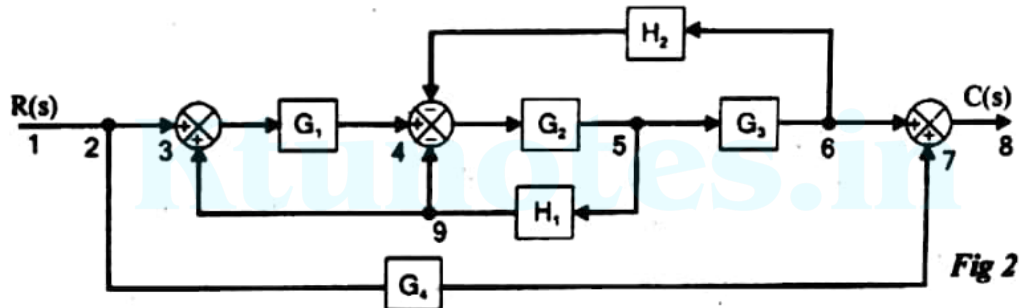
EXAMPLE 1.32

Convert the block diagram to signal flow graph and determine the transfer function using Mason's gain formula.

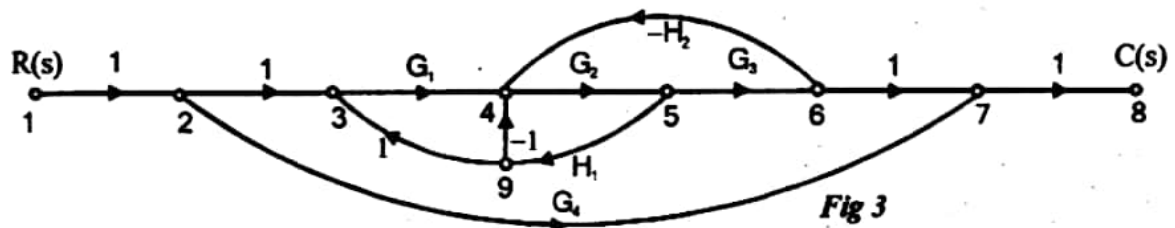


SOLUTION

The nodes are assigned at input, output, at every summing point & branch point as shown in fig 2.



The signal flow graph for the above block diagram is shown in fig 3.



Forward Path Gains

There are two forward path, $\therefore K=2$.

Let the forward path gains be P_1 and P_2 .

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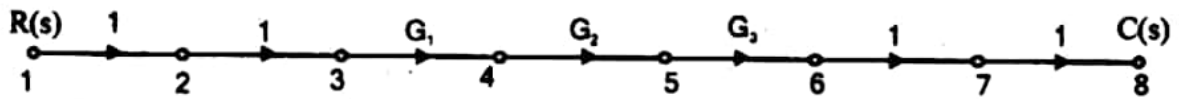


Fig 4 : Forward path-1.

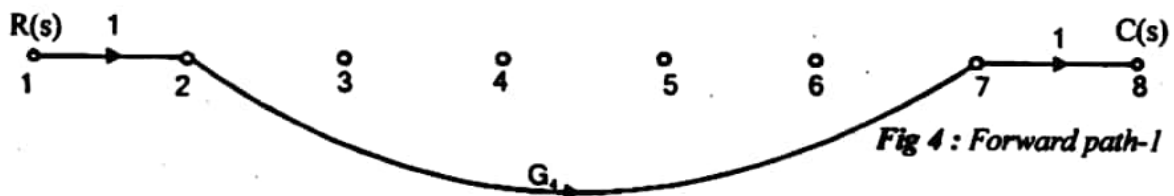


Fig 4 : Forward path-2

Gain of forward path-1, $P_1 = G_1 G_2 G_3$

Gain of forward path-2, $P_2 = G_4$

II. Individual Loop Gain

There are three individual loops with gains P_{11} , P_{21} and P_{31} .

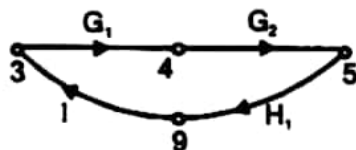


Fig 6 : loop-1.

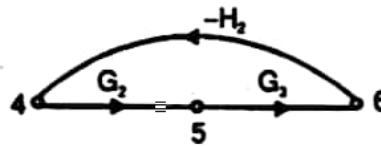


Fig 7 : loop-2.

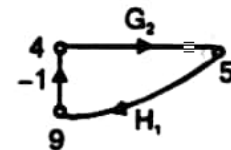


Fig 8 : loop-3.

II. Individual Loop Gain

There are three individual loops with gains P_{11} , P_{21} and P_{31} .

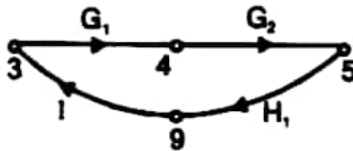


Fig 6 : loop-1.

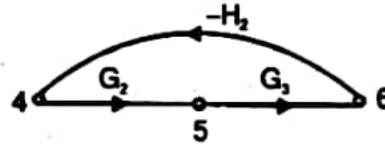


Fig 7 : loop-2.

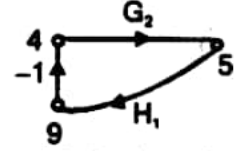


Fig 8 : loop-3.

Gain of individual loop-1, $P_{11} = G_1 G_2 H_1$

Gain of individual loop-2, $P_{21} = -G_2 G_3 H_2$

Gain of individual loop-3, $P_{31} = -G_2 H_1$

III. Gain Products of Two Non-touching Loops

There are no possible combinations of two non-touching loops, three non-touching loops, etc.,.

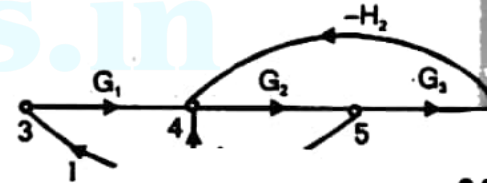
IV. Calculation of Δ and Δ_k

$$\Delta = 1 - [P_{11} + P_{21} + P_{31}] = 1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1$$

Since no part of graph touches forward path-1, $\Delta_1 = 1$.

The part of graph non touching forward path-2 is shown in fig 9.

$$\begin{aligned} \therefore \Delta_2 &= 1 - [G_1 G_2 H_1 - G_2 G_3 H_2 - G_2 H_1] \\ &= 1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1 \end{aligned}$$



V. Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

$$\begin{aligned} T &= \frac{1}{\Delta} \sum_k P_k \Delta_k = \frac{1}{\Delta} [P_1 \Delta_1 + P_2 \Delta_2] \quad (\text{Number of forward paths is 2 and so } K = 2) \\ &= \frac{1}{\Delta} [G_1 G_2 G_3 + G_4 (1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1)] \\ &= \frac{1}{\Delta} [G_1 G_2 G_3 + G_4 - G_1 G_2 G_4 H_1 + G_2 G_3 G_4 H_2 + G_2 G_4 H_1] \\ &= \frac{G_1 G_2 G_3 + G_4 - G_1 G_2 G_4 H_1 + G_2 G_3 G_4 H_2 + G_2 G_4 H_1}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_2 H_1} \end{aligned}$$

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EXAMPLE 1.33

Draw a signal flow graph and evaluate the closed loop transfer function of a system whose block diagram is shown in Fig 1.

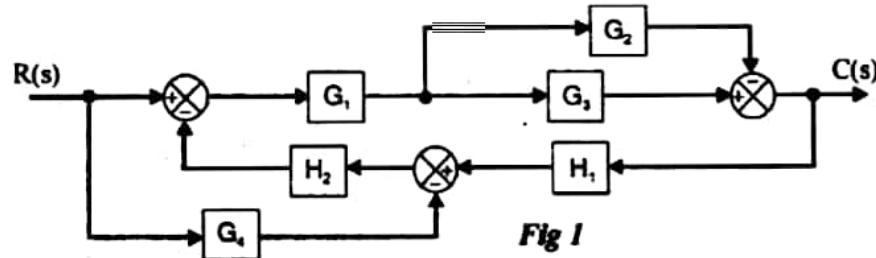


Fig 1

SOLUTION

The nodes are assigned at input, output, at every summing point & branch point as shown in Fig 2.

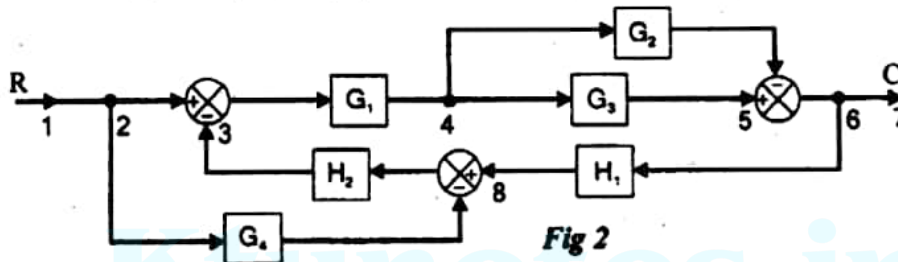


Fig 2

The signal flow graph for the block diagram of Fig 2, is shown in Fig 3.

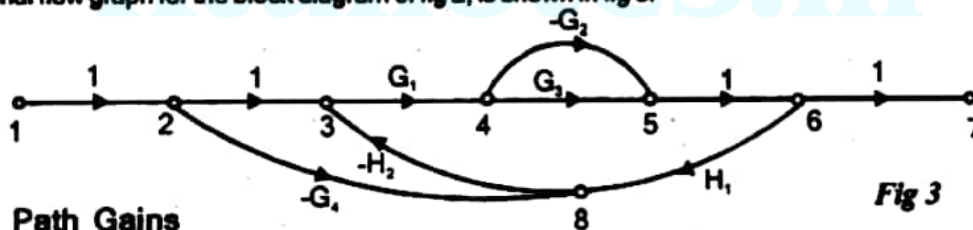


Fig 3

Forward Path Gains

There are four forward paths, $\therefore K = 4$

Let the forward path gains be P_1, P_2, P_3 and P_4 .

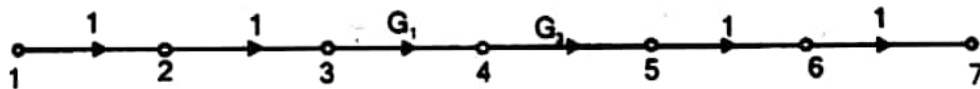


Fig 4 : Forward path-1.

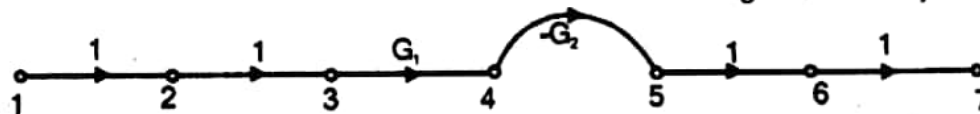


Fig 5 : Forward path-2.

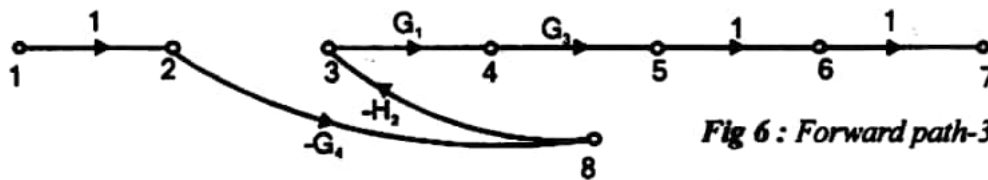
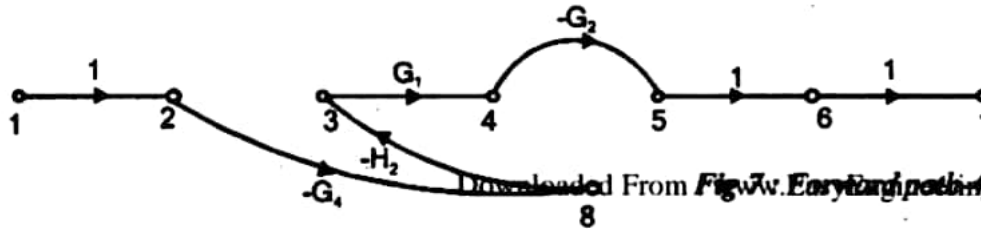


Fig 6 : Forward path-3.



Downloaded From Fig 7: Forward path-4.

1.91

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Gain of forward path-1, $P_1 = G_1 G_3$

Gain of forward path-2, $P_2 = -G_1 G_2$

Gain of forward path-3, $P_3 = G_1 G_3 G_4 H_2$

Gain of forward path-4, $P_4 = -G_1 G_2 G_4 H_2$

II. Individual Loop Gain

There are two individual loops, let individual loop gains be P_{11} and P_{21} .

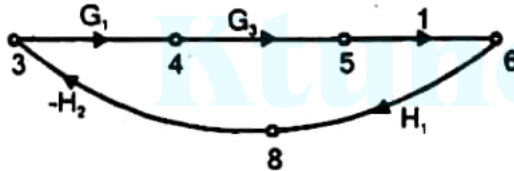


Fig 7 : loop-1

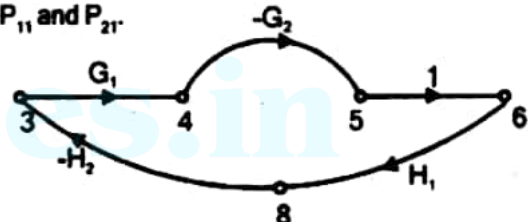


Fig 7 : loop-2

Loop gain of individual loop-1, $P_{11} = -G_1 G_3 H_1 H_2$

Loop gain of individual loop-2, $P_{21} = G_1 G_2 H_1 H_2$

III. Gain Products of Two Non-touching Loops

There are no possible combinations of two non-touching loops, three non-touching loops, etc..

IV. Calculation of Δ and Δ_k

$$\Delta = 1 - [\text{sum of individual loop gain}] = 1 - (P_{11} + P_{21})$$

$$= 1 - [-G_1 G_3 H_1 H_2 + G_1 G_2 H_1 H_2] = 1 + G_1 G_3 H_1 H_2 - G_1 G_2 H_1 H_2$$

Since no part of graph is non touching with the forward paths, $\Delta_1 = \Delta_2 = \Delta_3 = \Delta_4 = 1$.

V. Transfer Function, T

By Mason's gain formula the transfer function, T is given by,

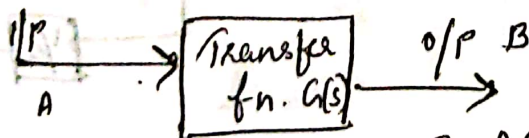
$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k = \frac{P_1 + P_2 + P_3 + P_4}{\Delta} \quad (\text{Number of forward paths is 4 and so } K = 4)$$

$$= \frac{G_1 G_3 - G_1 G_2 + G_1 G_3 G_4 H_2 - G_1 G_2 G_4 H_2}{1 + G_1 G_3 H_1 H_2 - G_1 G_2 H_1 H_2}$$

$$= \frac{G_1 (G_3 - G_2) + G_1 G_4 H_2 (G_3 - G_2)}{1 + G_1 H_1 H_2 (G_3 - G_2)} = \frac{G_1 (G_3 - G_2) (1 + G_4 H_2)}{1 + G_1 H_1 H_2 (G_3 - G_2)}$$

Block diagrams Representation Module - 3

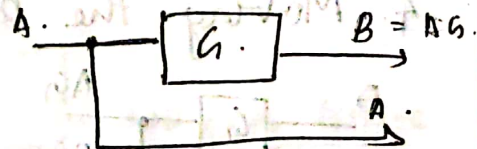
- A control s/m may consist of a no. of components.
- Block dgm is used to show the function of each component.
- A block diagram is a pictorial representation of the functions performed by each component.
- The elements of a block dgm - block, branch point and summing point.



$$B = A G(s).$$



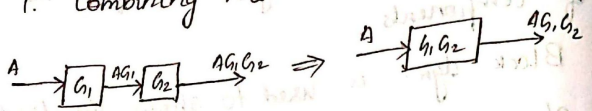
Summing pt - add two or more signal.



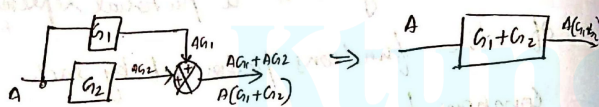
branch point
signal from one block goes to other blk

Rules of block diagram algebra.

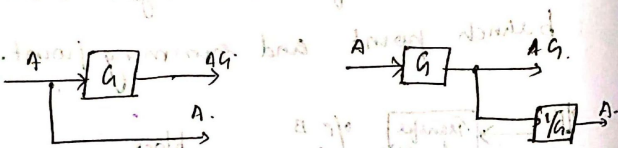
1. Combining the blocks in cascade.



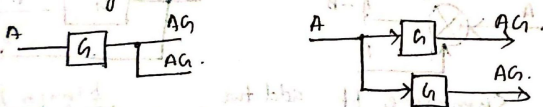
2. Combining parallel blocks.



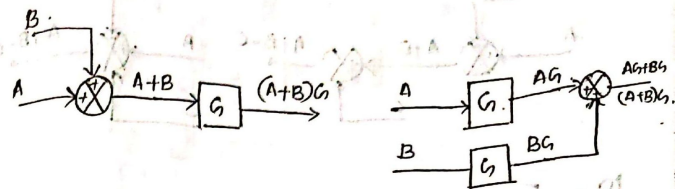
3. Moving the branch point ahead of the block.



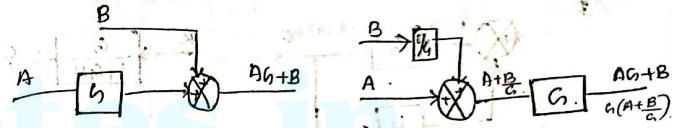
4. Moving the branch point before the block.



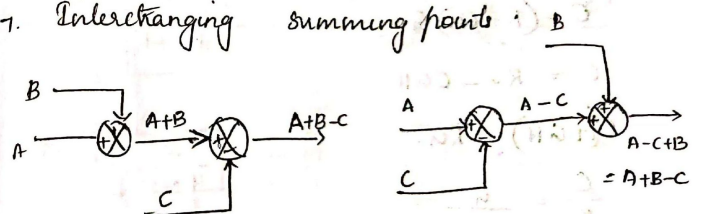
5. Moving the summing point ahead of the block.



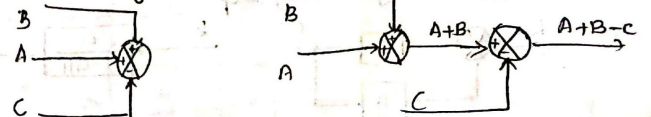
6. Moving the summing point before the block.



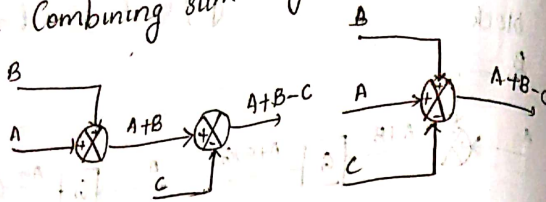
7. Interchanging summing points.



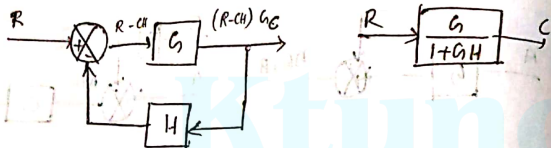
8. Splitting summing point.



9. Combining summing points



10. Feedback (negative) loop.



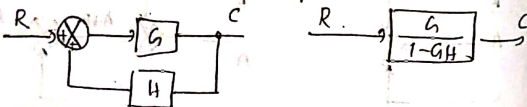
$$C = (R - CH) G_1$$

$$C = RG_1 - CG_1H$$

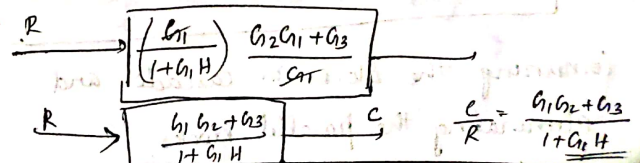
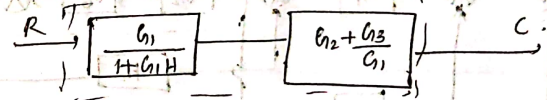
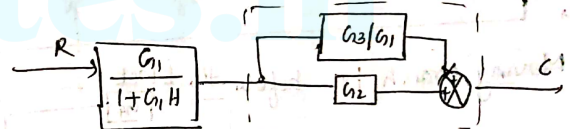
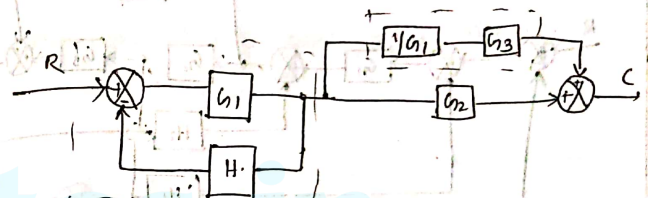
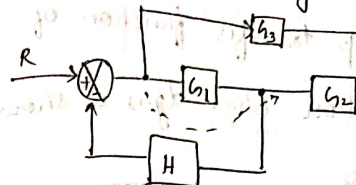
$$C(1 + G_1H) = RG_1$$

$$\frac{C}{R} = \frac{G_1}{1 + G_1H}$$

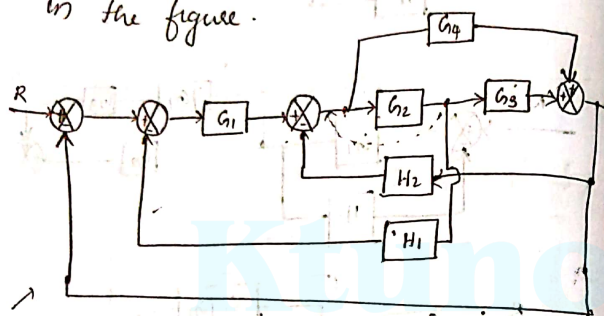
11. Feedback (positive) loop.



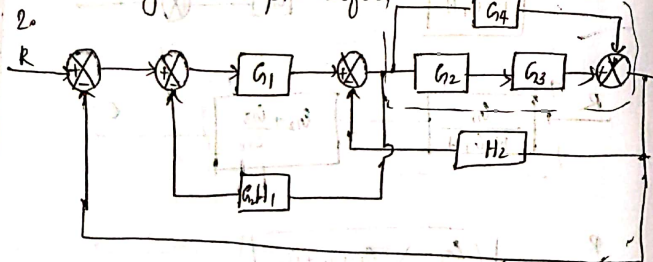
→ Reduce the block diagram shown in fig & find $\frac{C}{R}$



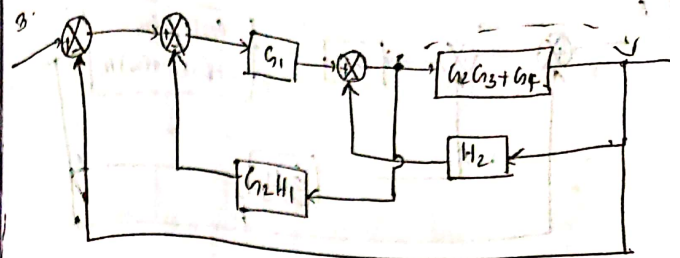
→ Using block diagram reduction technique find closed loop transfer function of the system whose block diagram is shown in the figure.



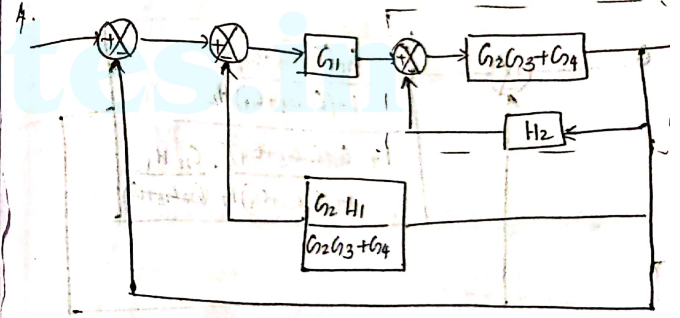
1. Moving branch pt. before the block.



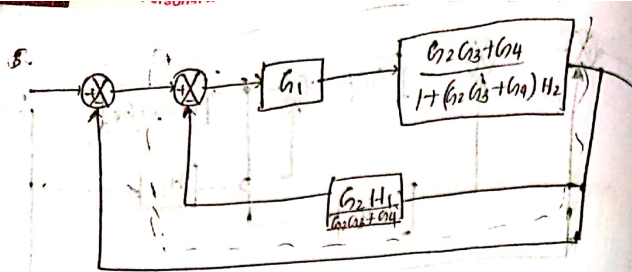
Combining the blocks in cascade and eliminating the parallel blocks.



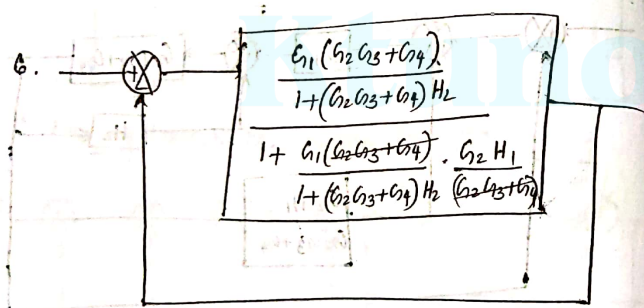
Moving the branch point after the block.



Eliminating the feedback loop.

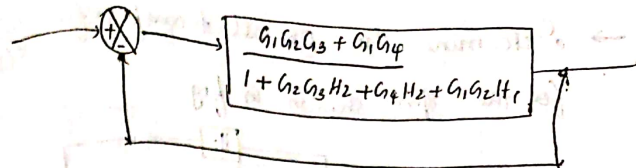


Combining the blocks in cascade and eliminating the feedback loop.



$$\frac{G_1 (G_2 G_3 + G_4)}{1 + (G_2 G_3 + G_4) H_2} \cdot \frac{G_2 H_1}{1 + (G_2 G_3 + G_4) H_2 + G_1 G_2 H_1 / (G_2 G_3 + G_4)}$$

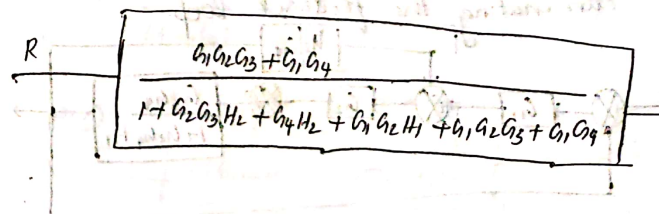
$$= \frac{G_1 (G_2 G_3 + G_4)}{1 + (G_2 G_3 + G_4) H_2 + G_1 G_2 H_1}$$



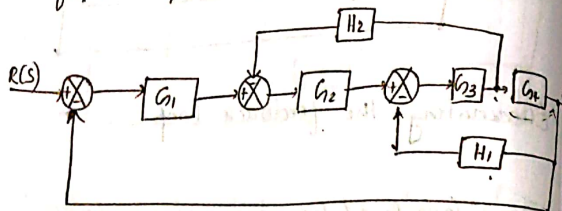
Eliminating the feedback loop

$$\frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_2 G_3 H_2 + G_4 H_2 + G_1 G_2 H_1}$$

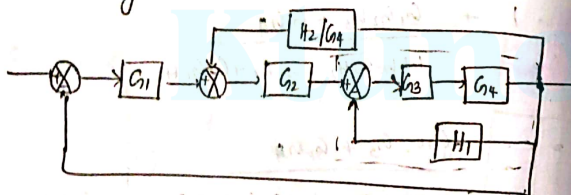
$$= \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_2 G_3 H_2 + G_4 H_2 + G_1 G_2 H_1 + G_1 G_2 G_3 + G_1 G_4}$$



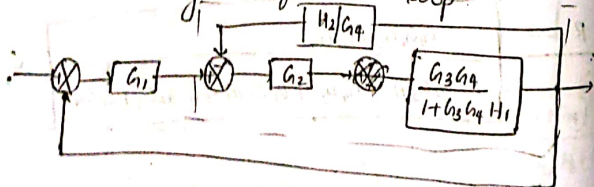
→ Determine the overall transfer for $\frac{C(s)}{R(s)}$ for the s/m shown in fig



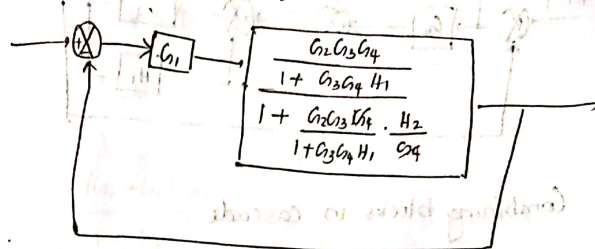
1. Moving the branch point after the block.



Eliminating the feedback loop



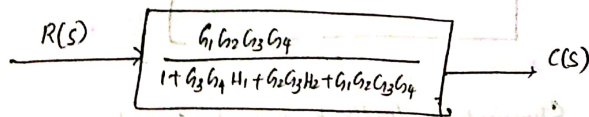
Eliminating the feedback loop.



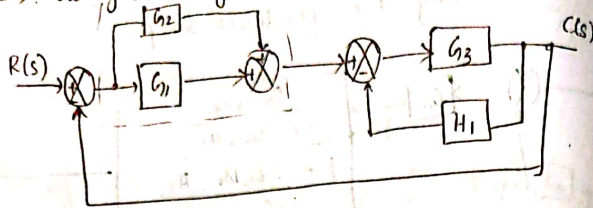
$$\frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2}$$

Eliminating f/b loop.

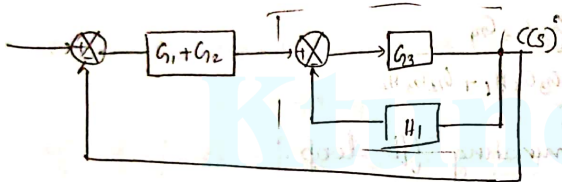
$$G_1 \cdot \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4}$$



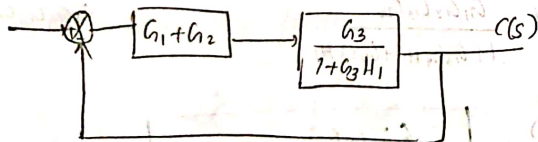
→ Using block diagram reduction, find the transfer function



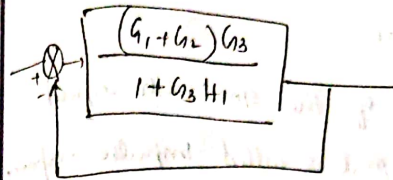
Combining blocks in cascade.



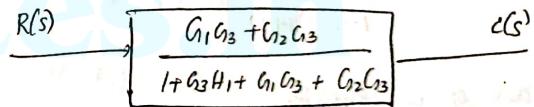
Eliminating feed back loop



Eliminating feedback loop & cascading



$$\frac{(G_1 + G_2) G_3}{1 + G_3 H_1} \div \left(1 + \frac{(G_1 + G_2) G_3}{1 + G_3 H_1} \right) = \frac{(G_1 + G_2) G_3}{1 + G_3 H_1 + G_1 G_3 + G_2 G_3}$$



Impulse response

The response of the s/m, with input as impulse signal is called impulse response.

- It is also given by inverse of Laplace transform of the s/m transfer function.

$$m(t) = L^{-1}\{R(s) M(s)\} = L^{-1}\{M(s)\}$$

$$M(s) = \frac{G(s)}{1+G(s)H(s)}$$

$R(s) = 1$ for impulse

Order of a system

$$T(s) = \frac{P(s)}{Q(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

The order of the system is given by the maximum power of s in the denominator polynomial $Q(s)$.

$\therefore$ Order is n .

$n=0 \rightarrow$ zero order s/m
 $n=1 \rightarrow$ first order s/m
 $n=2 \rightarrow$ second order s/m

$$T(s) = \frac{P(s)}{Q(s)} = \frac{(s+z_1)(s+z_2)\dots(s+z_m)}{(s+p_1)(s+p_2)\dots(s+p_n)}$$

$z_1, z_2, \dots, z_m \rightarrow$ zeros

$p_1, p_2, \dots, p_n \rightarrow$ poles

$\therefore$ Order = No. of poles.

Type number of system

If N is the no. of poles at the origin then type number is N .

Loop transfer fn, $G(s)H(s)$ is considered.

$$G(s)H(s) = K \cdot \frac{P(s)}{Q(s)} = K \cdot \frac{(s+z_1)(s+z_2)(s+z_3)\dots}{s^N (s+p_1)(s+p_2)\dots}$$

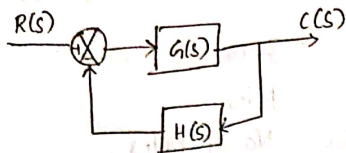
$N \rightarrow$ no. of poles at the origin

$N=0 \rightarrow$ type-0 s/m

$N=1 \rightarrow$ type-1 s/m

$N=2 \rightarrow$ type-2 s/m

Characteristic eq.



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)} \rightarrow \text{closed loop T.F.}$$

$$1 + G(s)H(s) = 0 \rightarrow \text{Characteristic eq. denominator of CLTF.}$$

Eg: $a_0 s^7 + a_1 s^6 + a_2 s^5 + a_3 s^4 + a_4 s^3 + a_5 s^2 + a_6 s + a_7 = 0$

Routh Hurwitz Criteria $\rightarrow$ stability of s/m.

Routh's array.

s^7	a_0	a_2	a_4	a_6
s^6	a_1	a_3	a_5	a_7
s^5	b_1	b_2	b_3	
s^4	c_1	c_2	c_3	
s^3	d_1	d_2		
s^2	e_1	e_2		
s^1	f_1			
s^0	g_2			

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

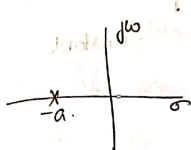
$$b_3 = \frac{a_1 a_6 - a_0 a_7}{a_1}$$

Check the first column of Routh array.

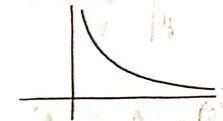
Location of poles on s-plane for stability

$$M(s) = \frac{A}{s+a}$$

$$m(t) = Ae^{-at}$$



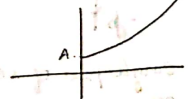
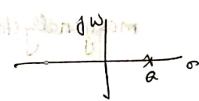
Root on -ve real axis



stable.

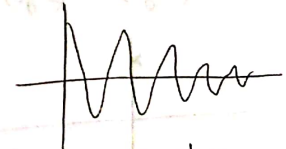
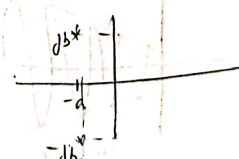
$$M(s) = \frac{A}{s-a}$$

$$m(t) = Ae^{at}$$



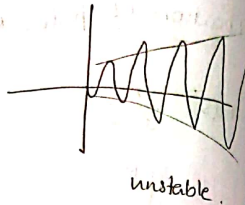
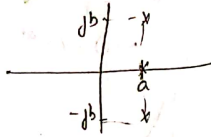
unstable.

$$M(s) = \frac{A}{s+a+jb} + \frac{A^*}{s+a-jb}$$

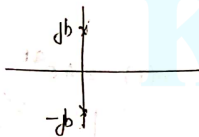


stable

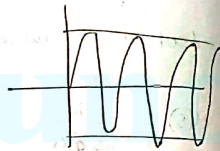
$$M(s) = \frac{A}{s-a+jb} + \frac{A^*}{s-a-jb}$$



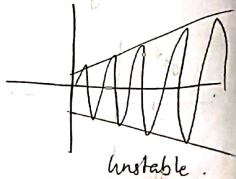
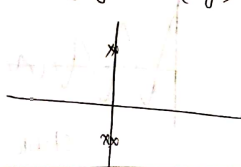
$$M(s) = \frac{A}{s+jb} + \frac{A^*}{s-jb}$$



single pair of roots on imaginary axis.



$$M(s) = \frac{A}{(s+jb)^2} + \frac{A^*}{(s-jb)^2}$$



1. If roots lie on negative half of s-plane, then s/m is stable.

2. If roots lie on positive half or have repeated roots on imaginary axis, then the system is unstable.

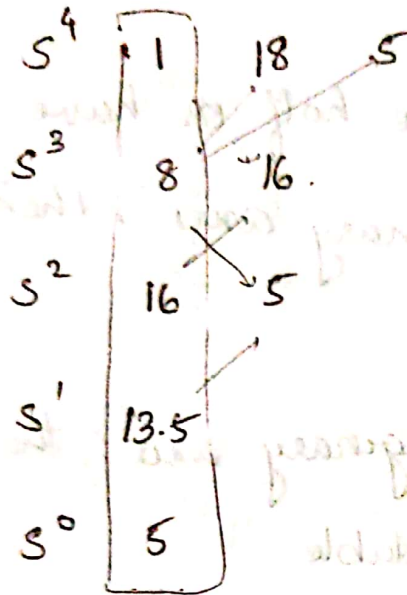
3. If roots lie on imaginary axis, then the s/m is marginally stable.

Routh Hurwitz Criterion

→ Using Routh Hurwitz criterion, determine the stability of the system represented by the characteristic eq. $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$.

Comment on the location of the roots of characteristic eq.

$$s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$$



$$\frac{8 \times 18 - 1 \times 16}{8} = 16$$

$$\frac{8 \times 5 - 0}{8}$$

$$\frac{8 \times 16 \times 16 - 8 \times 5}{16} = 16$$

All the elements of 1st column of Routh array are positive. Hence all the roots lie on the left half of s-plane and the system is stable.